QA TECHNICAL AUDIT

Test Report: Object Physics & Item Defects

Target Area: Interactive Prefabs & Collision Pipeline **Severity:** Critical / High **Engine:** Unity 2D

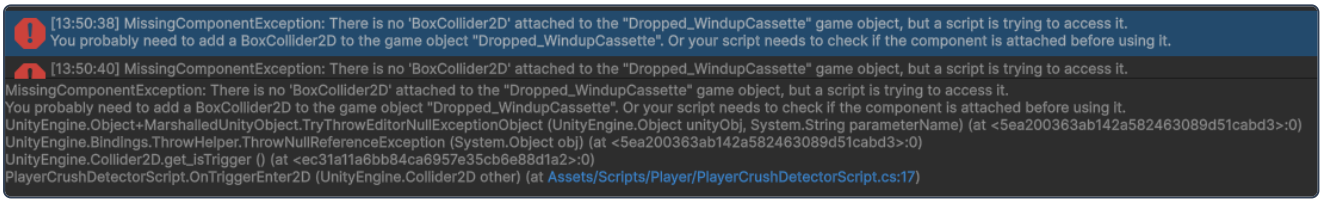
1. Detected Technical Bugs

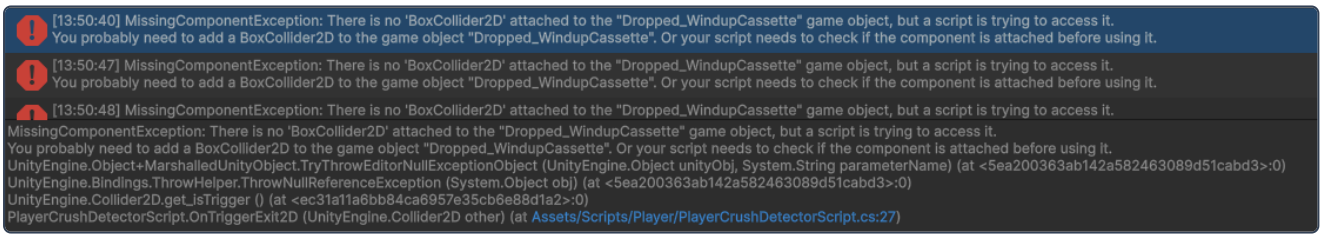
BoxCollider2D Component Error (MissingComponentException)

Upon interacting with dropped items, the engine triggers a critical runtime exception reporting the absence of a `BoxCollider2D` component. The script `PlayerCrushDetectorScript.cs` at lines 17 (`OnTriggerEnter2D`) and 27 (`OnTriggerExit2D`) attempts to access the collider of the dropped item. Because the component is missing on the instantiated object, an unhandled exception is thrown in the Unity console, compromising physics resolution and trigger detection.

> **Reproduction Path:** *Pick up an item (oil can or cassette) into the inventory → Drop the object onto the floor → Direct the player character into immediate physical contact/interaction with the dropped item. The bug reproduces consistently across both assets.*







Mesh Penetration & Intersection Defects (Clipping / Surface Bleed)

The player character can physically penetrate geometry or become wedged in unnatural orientations between static level geometry and the newly introduced piston/hazard prefabs. This behavior is provoked by inaccurate asset scale factors and improperly bounded trigger volumes.

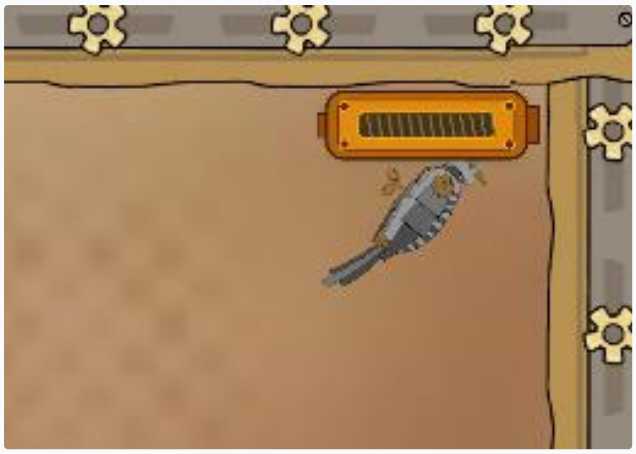


Figure 1: Player character trapped inside the upper piston housing

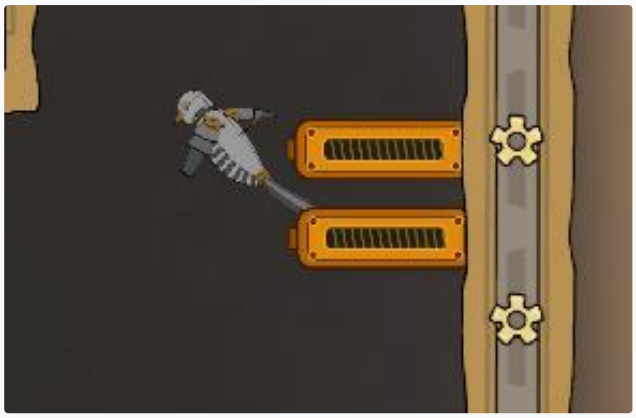


Figure 2: Inappropriate collision bounds causing character entrapment

2. Object Physics & Gameplay Experience (UX) Evaluation

Current mass and surface friction parameters configured on the newly introduced interactable items induce erratic physics responses and disrupt movement flow.

Object	Observed Physics Behavior	Gameplay Impact & Recommended Adjustments
Oil Bottle	The object has excessively low mass (overly lightweight). Upon the slightest bump or inadvertent contact while flying the cuckoo into it, the bottle is violently launched across large distances. The physics material exhibits negligible drag, causing excessive ground sliding.	**Player Frustration:** Brushing against the item without the intention of collecting it propels it down the entire hallway. **Remedy:** Increase `Mass` in `Rigidbody2D` and assign a `PhysicsMaterial2D` with higher friction to constrain lateral slide.
Cassette	Exhibits a comparable lack of perceived weight. The cassette responds unstably upon floor contact and glides without sufficient resistance across tile surfaces.	**Game Feel:** The item feels synthetic and weightless, degrading tactile immersion during physical handling. **Remedy:** Increase mass and gravity scale values in the physics component, accompanied by calibrated surface friction.

3. Developer Action Items (To-Do List)

- **BoxCollider2D Integration:** Ensure that both `Dropped_WindupCassette` and `Dropped_OilCan` prefabs have an active, properly configured `BoxCollider2D` component assigned.
- **Defensive Trigger Handling:** In `PlayerCrushDetectorScript.cs`, add explicit null-checks before invoking collider queries, e.g., `if (other.GetComponent<BoxCollider2D>() != null)`.
- **Rigidbody2D Recalibration:** Increase the `Mass` attribute and tune `Linear Drag` to suppress excessive rebound force and persistent surface sliding.